A Conditional Improvement of a Fixed Theta Operator Lower Bound

Jongmin Choi
Independent Researcher, Seoul, Korea
Email: 24ping@naver.com
ORCID: 0009-0008-7448-514X

Research note — 18 October 2026

Abstract

We record a small, conditional improvement of a fixed negative lower bound for the same minimum theta-weighted self-adjoint operator. An elementary shift-reuse inequality keeps a previously admitted trial map, its full-line action errors, all 372 observation coordinates and the original coefficient matrix. It needs no new source action, coefficient solve or full trial Gram matrix. We show first that scalar resolvent continuity using only the old coercivity and response margins gives exactly the old lower bound. A separately fixed shift beyond that scalar margin was then tested once with 68 trials. The complete independently implemented exact checker certified a fresh positive response margin and an improvement of approximately $1.29412611030898\cdot 10^{-5}$, from approximately -0.0156184544151869 to -0.0156055131540838. All analytic-source, operator-domain and physical-inclusion premises remain explicitly inherited. The new lower edge is negative. Nonnegativity, a sequence of thresholds tending to zero, and the Riemann hypothesis are not established. No bibliographical priority is claimed.

1 Purpose and the fixed operator

The object of this note is one fixed lower-bound improvement, not an unconditional reconstruction of the underlying theta certificate. The previously archived corrected calculation used 68 trials and all 372 observation coordinates [1]. We retain those objects and their coefficient matrix. The new work consists of a shift-reuse argument, a precise limitation of a scalar-only transfer, and one complete finite verification at a shift chosen before execution.

For $x \geq 0$, put

$$\kappa(x) = e^{x/2} \sum_{n\geq 1} (4z_n(x)^2 - 6z_n(x)) e^{-z_n(x)}, \qquad z_n(x) = \pi n^2 e^{2x}, \tag{1.1}$$

and extend κ evenly. Write

$$c(x) = \cosh(x/2), \qquad \mathrm{d}\nu(x) = 2\kappa(x)c(x)\,\mathrm{d}x, \qquad \mathcal{H} = L^2_{\mathrm{even}}(\mathbb{R}, \nu; \mathbb{C}). \tag{1.2}$$

The source normalization is $\nu(\mathbb{R}) = 1$. With Λ the von Mangoldt function, the symmetric half measure is

$$\mathrm{d}\mu(s) = \frac{1}{2}\frac{e^{-|s|/2}}{1-e^{-2|s|}}\,\mathrm{d}s + \frac{1}{2}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}(\delta_{\log n} + \delta_{-\log n}). \tag{1.3}$$

The value of the continuous density at zero is immaterial; its singularity is always used with the difference in the form. On the compact smooth even core, set

$$\mathcal{E}(f) = \iint \kappa(x)\kappa(z)|f(x) - f(z)|^2\,\mathrm{d}\mu(x-z)\,\mathrm{d}x. \tag{1.4}$$

The notation integrates z against the translated shift measure, including its atoms. Equivalently, the positive-shift integral has continuous weight $e^{-s/2}/(1-e^{-2s})$ and prime-power weight $\Lambda(n)/\sqrt{n}$, with no extra factor of two.

Let $(\mathcal{E},\mathcal{D})$ be the *minimum* closure of this core in the $\mathcal{E}+\mathcal{H}$ norm. Its representing operator is $A \geq 0$. We use the same operator throughout:

$$L = A - \frac{1}{2}I + \frac{1}{2}P_1, \qquad P_1 f = \langle 1, f\rangle_{\mathcal{H}}\, 1, \qquad \mathrm{D}(L) = \mathrm{D}(A). \tag{1.5}$$

Write ℓ for the associated form of L on $\mathcal{D}$. The inner product is conjugate-linear in its first variable. No equality of this minimum domain with a maximal jump-energy domain is assumed.

For orientation, the known source space is the closed span of 1 and $\kappa^{(2j)}/\kappa - 4^{-j}$, $j \geq 1$, in $\mathcal{H}$. The present argument does not assume that this space equals $\ker L$. Its lower bound holds on the whole form domain. Restricting it to $\mathcal{D} \cap \mathcal{N}^{\perp}$ does not constitute a new use of every source orthogonality condition to obtain coercivity.

1.1 Explicit inherited premises

The following premise labels retain their meaning from the source audit. They are not conclusions drawn from a stored PASS label.

Hypothesis 1.1 (Analytic construction A1–A4)**.** Assume:

(A1) The infinite source (1.1) is positive and even, holomorphic in $|\operatorname{Im} z| < \pi/4$, has Fourier transform Ξ in the source convention, and has the stated normalization and bounds.

(A2) The form (1.4) has the minimum closure just specified, with associated operator and domain (1.5).

(A3) The compact-core signed identity holds, and the complete sandwiched prime series converges in operator norm. Its omitted tail after $J \geq 7$ is bounded by $82\sum_{n>J} n^2 20^{-n}$.

(A4) The source multiplier cancellation, including the differentiated pole term, holds weakly against compact smooth tests. The stated locally uniform convergence and Gamma-decaying Mellin estimates justify each fixed number of differentiations.

Fix the archived observation synthesis map $O : \mathbb{C}^{372} \to \mathcal{H}$, in its original order and Cholesky coordinates, and put

$$\delta_0 = \frac{1}{64}, \qquad H_0 = L + \delta_0 I + OO^*. \tag{1.6}$$

There are 124 retained cells and three observation columns per cell. The trial map $V : \mathbb{C}^{68} \to \mathrm{D}(H_0)$ has the fixed order $V_0, \dots, V_{63}, q_0, q_1, q_2, q_3$. These four q_j are residual-source trials, not small-translation null vectors. They are true-source functions, not their compact saved action proxies. Their pole terms need not vanish at infinity.

Hypothesis 1.2 (Physical and numerical inclusions N1–N4)**.** Assume:

(N1) The whole-cell and source/moment intervals have their claimed inclusions, so $H_0 \geq \eta I$, where

$$\begin{aligned} \eta &= \frac{1}{64} - d_\sharp - \epsilon_{\mathrm{cover}} > 0, \\ d_\sharp &= \frac{960589}{4389760000000}, \\ \epsilon_{\mathrm{cover}} &= \frac{1419423310169286249010154329978348 1485}{3284055810247577849424937639623597936869376}. \end{aligned} \tag{1.7}$$

(N2) All 68 true trials lie in the minimum operator domain. The full-line proxy map $G : \mathbb{C}^{68} \to \mathcal{H}$ and $E = H_0V - G$ satisfy the admitted bounds

$$\|V\| \le \nu_0, \qquad \|E\| \le \bar{\sigma}, \qquad \|E\|^2 \le s^2. \tag{1.8}$$

Here s^2 is the saved sum of squared deterministic column bounds. In (1.8), $\|E\|^2$ means the square of the operator norm, not the norm of an operator composition.

(N3) The accepted interval Grams contain the full-line quantities O^*O, V^*O, $\mathrm{sym}(V^*G)$, G^*O, G^*G, with all documented integration, truncation, source-replacement, transfer and serialization errors.

(N4) The old exact postprocessing and factor data are the specified saved data. Their archived independent checker verifies the finite arithmetic conditionally on (N1)–(N3), without regenerating physical integrals.

The full affine exterior is part of (N2). In particular, for the first 64 columns, $G_{64} = G_{16} - (3/64)V_{64}$ on the whole line, including where the compact saved G_{16} proxy is zero. The four additional action proxies have their separately charged errors. The public `SOURCE_MAP.md` and frozen input-role manifest specify these definitions. This note neither re-evaluates their physical actions nor independently discharges Hypotheses 1.1–1.2.

2 A deterministic response bound and shift reuse

We first state a Hilbert-space argument independent of the arithmetic source. Let $H \ge aI > 0$ be self-adjoint, let $V : \mathbb{C}^r \to \mathrm{D}(H)$, $O : \mathbb{C}^m \to \mathcal{H}$, and $G = HV - E$. Assume the norm bounds (1.8). Fix any matrix $C : \mathbb{C}^m \to \mathbb{C}^r$ and any $\tau > 0$. No normal equation or optimality property is imposed on C.

Lemma 2.1 (Deterministic response majorant)**.** *Put $D = V^*O$, $F = \mathrm{sym}(V^*G)$, where $\mathrm{sym}\, X = (X + X^*)/2$, and*

$$\alpha = \frac{1+\tau}{a}, \qquad \beta = \nu_0\bar{\sigma} + \frac{(1+1/\tau)s^2}{a}.$$

Then, in Loewner order on $\mathbb{C}^m$,

$$O^*H^{-1}O \preceq C^*D + D^*C - C^*FC + \alpha(O - GC)^*(O - GC) + \beta C^*C. \tag{2.1}$$

Proof. Set $Y = VC$. Since Y takes values in $\mathrm{D}(H)$, expansion gives the exact finite-operator identity

$$\begin{aligned} O^*H^{-1}O = {} & O^*Y + Y^*O - Y^*HY \\ & + (O - HY)^*H^{-1}(O - HY). \end{aligned} \tag{2.2}$$

The inverse is bounded on all of $\mathcal{H}$; this identity requires no action of H on G or on O. Self-adjointness gives $V^*HV = F + \mathrm{sym}(V^*E)$. For $z \in \mathbb{C}^m$,

$$|\langle VCz, ECz\rangle| \le \nu_0\bar{\sigma}\|Cz\|^2.$$

Writing $R = O - GC$, deterministic Young's inequality gives

$$\|(R - EC)z\|^2 \le (1+\tau)\|Rz\|^2 + (1+1/\tau)s^2\|Cz\|^2.$$

Use $H^{-1} \preceq a^{-1}I$ in (2.2) and combine these estimates. □

For the fixed H_0, V, G, O, write

$$M = O^*O, \quad D = V^*O, \quad F = \mathrm{sym}(V^*G), \quad K = G^*O, \quad N = G^*G, \quad P = V^*V. \tag{2.3}$$

The archive roles for M, D, F, K, N are respectively `A,D68,E68,K68,N68`. In particular, `E68` does not denote the action error E.

Lemma 2.2 (Shift reuse with fixed coefficients)**.** *Let $0 < h < \eta$, and keep O, V, C fixed. Set*

$$H_h = H_0 - hI, \qquad \eta_h = \eta - h, \qquad G_h = G - hV,$$

and define

$$\alpha_h = \frac{1+\tau}{\eta_h}, \qquad \beta_h = \nu_0\bar{\sigma} + \frac{(1+1/\tau)s^2}{\eta_h}, \tag{2.4}$$

$$B_h = \alpha_h K - (1+\alpha_h h)D, \tag{2.5}$$

$$\overline{S}_h = \alpha_h N - (1+2\alpha_h h)F + [\beta_h + (h+\alpha_h h^2)\nu_0^2]I_r, \tag{2.6}$$

$$\overline{U}_h(C) = \alpha_h M - B_h^* C - C^* B_h + C^* \overline{S}_h C. \tag{2.7}$$

Then

$$O^* H_h^{-1} O \preceq \overline{U}_h(C). \tag{2.8}$$

The same full-line action error E is used, and no entrywise enclosure of P or second action is required.

Proof. A bounded scalar perturbation leaves both the operator and form domains unchanged. We have $H_h \geq \eta_h I$ and, on the fixed admitted trials,

$$H_h V - G_h = (H_0 - hI)V - (G - hV) = E. \tag{2.9}$$

The shifted Grams satisfy exact identities

$$F_h = F - hP, \qquad K_h = K - hD, \qquad N_h = N - 2hF + h^2 P.$$

Apply Lemma 2.1 with H_h, G_h, η_h and expand. The result is (2.7) with $\overline{S}_h$ replaced by

$$S_h = \alpha_h N - (1+2\alpha_h h)F + (h+\alpha_h h^2)P + \beta_h I_r.$$

Since $P \preceq \nu_0^2 I_r$ and $h + \alpha_h h^2 > 0$, this replacement by $\overline{S}_h$ increases the majorant. No positivity of F, S_h or $\overline{S}_h$ is needed. □

The norm replacement may lose margin. The lemma proves validity, not sharpness or successful closure. Its proof is an elementary residual identity and norm estimate; no literature-priority claim is made. The unchanged C is a legitimate trial coefficient matrix without a new solve or a separate optimal-coefficient theorem.

3 Why scalar continuity alone does not improve the edge

Suppose the old full-coordinate certificate gives

$$O^* H_0^{-1} O \preceq (1-\mu_0) I_m, \qquad 0 < \mu_0 < 1, \qquad \delta_* = \mu_0 \eta. \tag{3.1}$$

It implies the old lower edge $b_0 = -\delta_0 + \delta_*$.

Proposition 3.1 (No gain from the two old scalars)**.** *For $0 \le h < \eta$, spectral calculus gives*

$$I_m - O^* H_h^{-1} O \succeq \frac{\delta_* - h}{\eta - h} I_m. \tag{3.2}$$

For $0 \le h \le \delta_*$, the lower bound obtained by combining this margin with $H_h \ge (\eta - h)I$ is exactly $L \ge b_0 I$. For $h > \delta_*$, (3.2) supplies no positive response margin.

Proof. For $\lambda \ge \eta$,

$$(\lambda - h)^{-1} \le \frac{\eta}{\eta - h} \lambda^{-1}.$$

Functional calculus followed by (3.1) proves (3.2). For the nonnegative margin $m_h = (\delta_* - h)/(\eta - h)$, the form argument in Proposition 4.2 below yields

$$L + (\delta_0 - h)I \ge m_h(\eta - h)I = (\delta_* - h)I.$$

The h terms cancel, giving b_0. At $h = \delta_*$, the response margin is zero. Beyond it, this particular scalar estimate cannot certify positivity. This does not assert that the actual response matrix has lost positivity. □

4 From finite intervals to the original form domain

The arithmetic implementation encloses (2.7) by signed outward interval operations, retaining inherited physical radii and new rounding radii. In particular, the negative coefficient of F reverses endpoints. Suppose the resulting Hermitian interval matrix has symmetric center U_c and symmetric nonnegative radii r_{ij}. Define

$$U_{\mathrm{cert}} = U_c + \operatorname{diag}\Big(\sum_j r_{ij}\Big). \tag{4.1}$$

For any Hermitian error Δ in these radii,

$$z^* \Delta z \le \sum_{ij} r_{ij} |z_i| |z_j| \le \sum_i \Big(\sum_j r_{ij}\Big) |z_i|^2.$$

Thus $U_{\mathrm{cert}} \succeq \overline{U}_h(C)$. An entrywise upper-endpoint matrix alone would not establish this Loewner bound.

For $T = I_m - U_{\mathrm{cert}}$, the saved witness consists of a unit-lower dyadic matrix L_f, a positive diagonal D_f, and an approximate inverse factor M_f. Suppose exact saved-data checks establish

$$\begin{gathered} \|T - L_f D_f L_f^*\|_2 \le \rho, \qquad \|L_f M_f - I\|_1 \le e_1 < 1, \quad \|L_f M_f - I\|_\infty \le e_\infty < 1, \\ \|M_f\|_1 \le m_1, \quad \|M_f\|_\infty \le m_\infty, \qquad d = \min_i (D_f)_{ii} > 0. \end{gathered} \tag{4.2}$$

The symmetric factor residual can be bounded by its maximum absolute row sum.

Lemma 4.1 (Finite positivity witness)**.** *Under (4.2), define*

$$\mu_1 = \frac{d(1 - e_1)(1 - e_\infty)}{m_1 m_\infty} - \rho. \tag{4.3}$$

If $\mu_1 > 0$, then $T \succeq \mu_1 I_m$.

Proof. The Neumann bounds in the 1- and ∞-norms and $L_f^{-1} = M_f (L_f M_f)^{-1}$ give

$$\|L_f^{-1}\|_2^2 \le \|L_f^{-1}\|_1 \|L_f^{-1}\|_\infty \le \frac{m_1 m_\infty}{(1 - e_1)(1 - e_\infty)}.$$

Consequently $L_f D_f L_f^* \succeq (\mu_1 + \rho) I_m$. Subtract the residual bound in (4.2). □

Proposition 4.2 (Form-domain transfer)**.** *If $H_h \geq \eta_h I > 0$ and $O^* H_h^{-1} O \preceq (1-\mu) I_m$, where $0 \leq \mu \leq 1$, then*

$$L + (\delta_0 - h) I = H_h - OO^* \geq \mu H_h \geq \mu \eta_h I \tag{4.4}$$

as forms on the original $\mathcal{D}$.

Proof. Let $B = H_h^{-1/2} O$. Then $\|B\|^2 = \|B^* B\| \leq 1 - \mu$. For every $f \in \mathcal{D} = \mathrm{D}(H_h^{1/2})$,

$$\|O^* f\|^2 = \|B^* H_h^{1/2} f\|^2 \leq (1-\mu) \|H_h^{1/2} f\|^2.$$

Subtract this bounded form from that of H_h, and use $\mu \geq 0$. There is no compression of the conclusion to the trial span. □

5 The one fixed shifted certificate

The original archived margin satisfies $0 < \mu_0 < 1/100$, and $0 < \eta < \delta_0$. Before execution the candidate was fixed as

$$h = 2\delta_* = 2\mu_0 \eta, \qquad \delta_1 = \delta_0 - h > 0, \qquad \eta_1 = \eta - h > 0, \qquad \tau = \frac{1}{100}. \tag{5.1}$$

The 68-by-372 matrix C was copied byte for byte from the previous certificate. No coefficient solve, candidate search or change of precision was allowed.

Theorem 5.1 (Conditional fixed-threshold improvement)**.** *Assume Hypotheses 1.1–1.2 and the exact inclusions and finite witness checked in the public certificate specified in Appendix A. Let μ_1 be its exact rational margin. Then on the original form domain,*

$$\ell(f, f) \geq b_1 \|f\|^2 \quad (f \in \mathcal{D}), \qquad b_1 = -\delta_0 + 2\delta_* + \mu_1(\eta - 2\delta_*), \tag{5.2}$$

where ℓ is the associated form of L. Moreover,

$$b_1 - b_0 = \delta_* + \mu_1(\eta - 2\delta_*) > 0, \qquad b_1 < 0. \tag{5.3}$$

The exact $\mu_1, b_0, b_1, b_1 - b_0$ are the fields `mu1`, `previous_lower_edge`, `new_lower_edge`, `strict_improvement` *of the commit-pinned* `EXACT_RESULT_VALUES.json`.

Proof. The complete interval assembly and Lemma 2.2 give $O^* H_h^{-1} O \preceq U_{\mathrm{cert}}$. The checked witness and Lemma 4.1 give $I_{372} - U_{\mathrm{cert}} \succeq \mu_1 I_{372}$, with $0 < \mu_1 < 1$. Proposition 4.2 yields (5.2). Subtracting $b_0 = -\delta_0 + \delta_*$ gives the exact identity in (5.3); both terms are positive. The recorded exact numerator of b_1 is negative and its denominator positive. □

The rational strings in the public JSON were generated inside the charged execution and copied verbatim. For readability, Table 1 formats those strings to 18 significant digits. The rounded values are not used to decide a sign or prove an inequality.

The improvement is strict but small in absolute size. It is not a claim of progress toward proving nonnegativity: the certified edge remains negative, the unchanged cover has the positive error floor $d_\sharp + \epsilon_{\mathrm{cover}}$, and no family with edges tending to zero has been supplied. The result does not determine whether L actually has negative spectrum.

Table 1: Display-only decimals of the saved exact rational quantities.

Quantity	Rounded display
Fresh margin μ_1	0.000409785708624514505
Fixed shift h	0.000013091169626178421 9
δ_1	0.0156119088303738216
η_1	0.0156073678398113979
Old edge b_0	−0.0156184544151869108
New edge b_1	−0.0156055131540838210
Improvement $b_1 - b_0$	0.0000129412611030897828

5.1 What the computation checked

The fixed run evaluated one full interval consumer with 160-bit input/scalar rounding, the original 80-bit C, and 100-bit final model rounding. It made one fixed-order block-16 factorization and one finite triangular inverse, with 80-bit witness serialization and a 1024-bit dense-integer ceiling. All physical error pads remained in the majorant.

The independently implemented exact checker verified all $372^2 = 138{,}384$ model locations using 69,378 independent triangular computations, all 372 radius row sums, the signed shifted model, all $64 + 4$ action-error bounds and norm majorants, unchanged C, the factor residual, inverse-factor norms and the strict-improvement identity. It imported neither producer helpers nor native arithmetic and performed no factorization or solve. The producer's native runtime nevertheless remained mapped in that same worker. This is implementation independence, not process isolation.

The producer used python-flint 0.9.0 and FLINT 3.6.0. Midpoint-radius arithmetic provides enclosures for the operations performed [2]; it does not establish that physical input intervals contain the mathematical objects asserted in Hypothesis 1.2. Those inclusion premises remain explicit.

5.2 Resource evidence and distinct earlier failure

The successful launch identity was `theta-full372-h2-star-fixedC-20261018-compat01`. It ran once with no scientific retry, coefficient solve, new source action, physical quadrature, alternate candidate or precision upgrade. Runtime/ABI admission occurred inside that execution.

The external host bracket, beginning before its first fork and ending after actual observer reaping and final `ECHILD`, was 20,839,672,121 nanoseconds, or 20.839672121 seconds. All registered roles were restricted to one CPU, so this also bounds their aggregate CPU capacity. It is not a separately measured total CPU usage. The limit was 60 seconds. The worker address-space *limit* was 384 MiB; this is not a measured peak. Its observed RSS/HWM maximum was 166,520 KiB and its final `wait4` maximum RSS was 166,348 KiB, both below the 320-MiB cap. The actual host tool exited zero; the worker, controller and observer exited zero, and the registered process tree was closed.

These measurements concern that calculation only. Research preparation, static review, manuscript work, compilation, upload and readback are outside the quoted interval. The process audit assumes the recorded trusted host/OS/source boundary and is not a malicious-native-code sandbox.

A distinct preceding host command exited one before its first fork because a `/proc/.../children` path was unavailable. No scientific worker or matrix calculation ran in that failed launch. A separately reviewed registered-tree implementation, a nonmathematical lifecycle fixture and a new fixture-bound release preceded the successful identity above. That distinct failure is archived separately and is not a numerical counterexample.

6 Reproducibility and limits of the claim

The public package contains the 63 frozen input roles (62 distinct files), the executed source, plan and command records, all saved mathematical outputs, the checker report, resource/closure receipts and actual tool transcript. The final timed manifest binds 29 files, and the checker binds 16 mathematical outputs. File hashes establish identity, not truth of a premise.

The package distinguishes read-only byte/binding verification from optional reader-side saved arithmetic replay and from full environment reconstruction. The unchanged checker and its original absolute-root admission are preserved. An unexecuted reader adapter and explicit filesystem/environment instructions are supplied; no newly tested hermetic runtime is claimed. Native runtime payloads are referenced through the prior public archive and their recorded hashes rather than redistributed in this note.

Paper preparation performed byte checks, formula/text review, scalar decimal formatting and document compilation. It did not rerun the matrix producer or checker, recompute physical inputs, refactor a matrix or solve for new coefficients. The original resource-pending status in the timed checker report is preserved; the later external witness and actual host exit resolve that condition.

The abstract argument is proved above. The application remains conditional on the stated source, domain and physical inclusions. Internal reviews and separate checker implementations are not external academic peer review or formal proof-assistant verification. No assertion of bibliographical novelty, full-nullspace completeness, $L \geq 0$, or the Riemann hypothesis is made. The recorded status remains `RH_PROOF=NOT-OBTAINED`.

A Commit-pinned public evidence

The source and data commit cited by this manuscript is

`dba8f9c497305eeb44fc4590604874480f5d256d`.

The complete artifact directory is publicly available at

https://github.com/serabing-hash/theta-weighted-certificate/tree/dba8f9c497305eeb44fc4590604874480f5d256d/research/2026-10-18-fixed-threshold-shift

This pins the data/source commit, rather than a moving branch or the later paper commit. The exact JSON is reached by the commit-pinned exact-result link; the timed checker report and external terminal witness are pinned to the same commit.

The exact-result file SHA-256 is

`08d52ed1f96eaa5b987214c073625e283adad22349c67c01d6a22c344a6cfe25`

and the original input-manifest SHA-256 is

`5bb28142e4bbc8fee7060467097db8b732fceb1564857b5dcca5ef3326b3fc58`.

The unchanged C has SHA-256

`c459b5c283020808e6d52ac60e3749137c11cd9db21e98fa5178ff0698ade513`.

The timed checker source has SHA-256

`e6ba2e463ea098276a99ea03175051ff2b9b5cbce09b2e602edf1387bc5b2a2f`.

The package's `MANIFEST.sha256`, `PUBLIC_ALLOWLIST.json`, `SOURCE_MAP.md`, `ASSUMPTIONS.md` and `REPRODUCTION.md` provide the remaining identities and execution limits.

References

[1] J. Choi, *Fixed-threshold source audit and corrected full372 public evidence*, research archive, 8 October 2026. https://github.com/serabing-hash/theta-weighted-certificate/releases/tag/v2026.10.08-fixed-threshold-source-audit. The archive is used with its explicit analytic and numerical inclusion premises.

[2] F. Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, IEEE Transactions on Computers **66** (2017). Author's preprint: https://arxiv.org/abs/1611.02831.

© 2026 Jongmin Choi. The author's original contributions are licensed under CC BY-NC 4.0, following the repository license. Third-party materials retain their original terms.